Advanced Counting Techniques

Chapter 8

Applications of Recurrence Relations

Section 8.1

Video 67: Counting with Recurrence Relations

Counting with Recurrence Relations

- Counting Bitstrings
- The Tower of Hanoi
- Counting Arrangments of Parantheses

Recurrence Relations

Definition: A **recurrence relation** for the sequence $\{an\}$ is an equation that expresses a_n in terms of a finite number k of the preceding terms of the sequence, i.e.,

$$a_n = f(a_{n-1}, a_{n-2,} \dots, a_{n-k})$$

A sequence $\{an\}$ is called a **solution** of a recurrence relation if its terms satisfy the recurrence relation.

The **initial conditions** for a sequence specify the terms $a_0, a_1, \dots, a_{k-1}$

Fibonacci Sequence

Definition: The **Fibonacci sequence** $f_0, f_1, f_2, \ldots$ is defined as:

Initial Conditions: $f_0 = 0, f_1 = 1$

Recurrence Relation: $f_n = f_{n-1} + f_{n-2}$

$f_2 = f_1 + f_0 = 1 + 0 = 1$

$f_3 = f_2 + f_1 = 1 + 1 = 2$

$f_4 = f_3 + f_2 = 2 + 1 = 3$

$f_5 = f_4 + f_3 = 3 + 2 = 5$

$f_6 = f_5 + f_4 = 5 + 3 = 8$

Counting Bit Strings

Find a recurrence relation and give initial conditions for the number of bit strings of length n without two consecutive 0s.

Let A_n denote the bit strings of length $n \geq 3$ without two consecutive 0s and let $a_n = |A_n|$

A bitstring $s_n \in A_n$ can end in a 1 or a 0, i.e. $s_n = s_{n-1}1$ or $s_n = s_{n-1}0$ for some $s_{n-1} \in A_{n-1}$

- If s_n ends in a 1, $s_n = s_{n-1}$, where s_{n-1} can be any bitstring from A_{n-1}
- If s_n ends in a 0, the s_{n-1} has to end in a 1; thus $s_{n-1} = s_{n-2}1$ where where s_{n-2} can be any bitstring from A_{n-2}

Therefore $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$.

Counting Bit Strings

How many such bit strings are there of length five?

The initial conditions are:

- $a_1 = 2$, since both the bit strings 0 and 1 do not have consecutive 0s.
- $a_2 = 3$, since the bit strings 01, 10, and 11 do not have consecutive 0s, while 00 does.

To obtain a_5, we use the recurrence relation three times to find that:

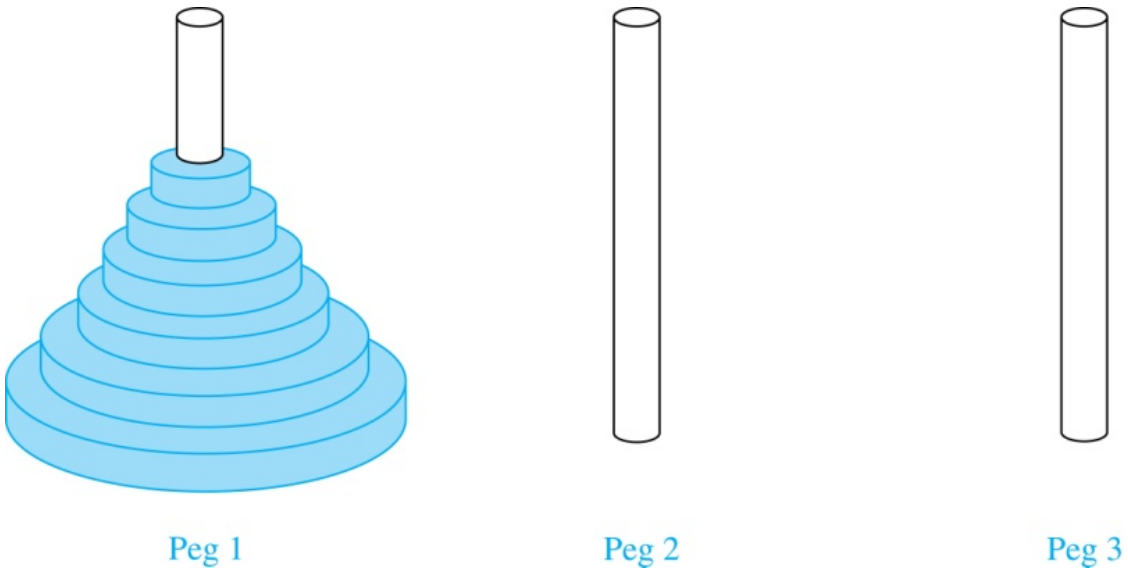
- $a_3 = a_2 + a_1 = 3 + 2 = 5$
- $a_4 = a_3 + a_2 = 5 + 3 = 8$
- $a_5 = a_4 + a_3 = 8 + 5 = 13$

Note that $\{a_n\}$ satisfies the same recurrence relation as the Fibonacci sequence.

The Tower of Hanoi

A puzzle consisting of three pegs on a board with disks of different sizes.

Initially all of the disks are on the first peg in order of size, with the largest on the bottom.



The Tower of Hanoi - Rules

Rules: You are allowed to move the disks one at a time from one peg to another as long as a larger disk is never placed on a smaller.

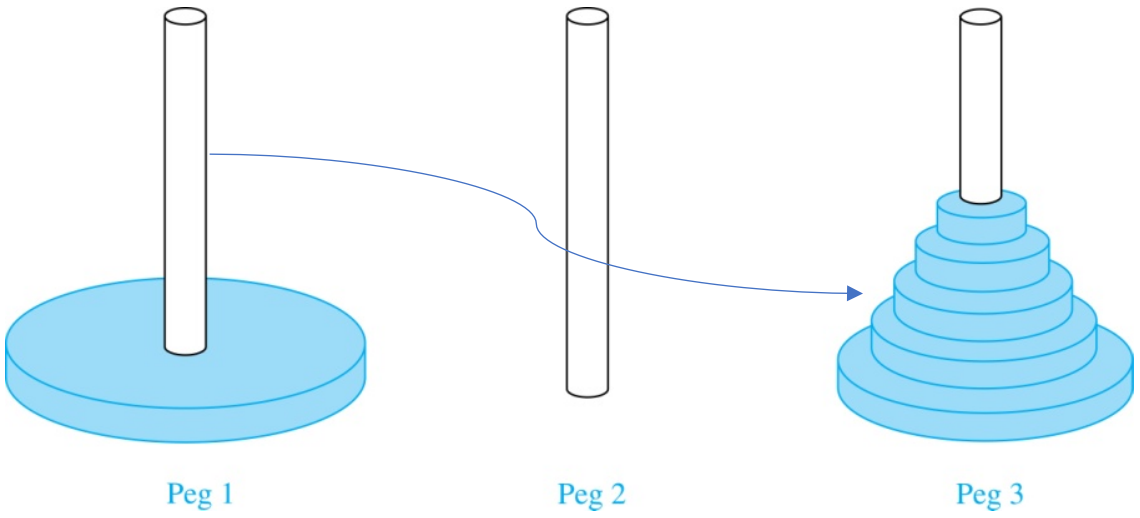
Goal: Using allowable moves, end up with all the disks on the second peg in order of size with largest on the bottom.

Question: How many moves are needed?

Let $\{H_n\}$ denote the number of moves needed to solve the Tower of Hanoi Puzzle with n disks.

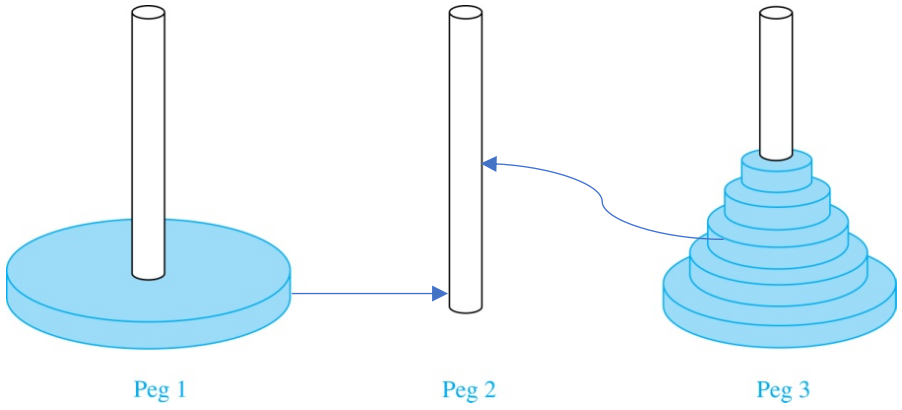
The initial condition is $H_1 = 1$ since a single disk can be transferred from peg 1 to peg 2 in one move.

The Tower of Hanoi – Initial Move



Begin with n disks on peg 1. We can transfer the top $n-1$ disks, following the rules of the puzzle, to peg 3 using H_{n-1} moves.

The Tower of Hanoi – Recursion



Then, we use 1 move to transfer the largest disk to the second peg.

Then we transfer the $n-1$ disks from peg 3 to peg 2 using H_{n-1} additional moves.

Hence, $H_n = 2H_{n-1} + 1$.

The Tower of Hanoi – Solving the Recurrence

$$
\begin{aligned}
H_n &= 2H_{n-1} + 1 \\
&= 2\,(2H_{n-2} + 1) + 1 = 2^2\, H_{n-2} + 2 + 1 \\
&= 2^2\,(2H_{n-3} + 1) + 2 + 1 = 2^3\, H_{n-3} + 2^2 + 2 + 1 \\
&\vdots \\
&= 2^{n-1}\, H_1 + 2^{n-2} + 2^{n-3} + \ldots + 2 + 1 \\
&= 2^{n-1} + 2^{n-2} + 2^{n-3} + \ldots + 2 + 1 \\
&= 2^n - 1
\end{aligned}
$$

since $H_1 = 1$

sum of the terms of a geometric series

$$\sum_{j=0}^{n} ar^j = \begin{cases} \frac{ar^{n+1}-a}{r-1} & r \neq 1 \\ (n+1)a & r = 1 \end{cases}$$

The Tower of Hanoi - History

There was a myth created around the puzzle:

Monks in a tower in Hanoi are transferring 64 golden disks from one peg to another following the rules of the puzzle. They move one disk each day. When the puzzle is finished, the world will end.

Using the closed formula for the 64 golden disks

$$2^{64} - 1 = 18{,}446{,}744{,}073{,}709{,}551{,}615$$

days are needed to solve the puzzle, which is more than 500 billion years.

Counting the Ways to Parenthesize a Product

Find a recurrence relation for C_n, the number of ways to parenthesize the product of $n + 1$ numbers, $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_n$, to specify the order of multiplication.

For example, $C_3 = 5$, since all the possible ways to parenthesize 4 numbers are

$((x_0 \cdot x_1) \cdot x_2) \cdot x_3$

$(x_0 \cdot (x_1 \cdot x_2)) \cdot x_3$

$(x_0 \cdot x_1) \cdot (x_2 \cdot x_3)$

$x_0 \cdot ((x_1 \cdot x_2) \cdot x_3)$

$x_0 \cdot (x_1 \cdot (x_2 \cdot x_3))$

Counting the Ways to Parenthesize a Product

Note that however parentheses are inserted in $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_n$,
one multiplication operator remains outside all parentheses.

- This final operator appears between two of the $n + 1$ numbers, say x_k and x_{k+1}.
- There are C_k ways to insert parentheses in the product $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_k$
- There are C_{n-k-1} ways to insert parentheses in the product $x_{k+1} \cdot x_{k+2} \cdot \cdots \cdot x_n$

Therefore we have

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + \cdots + C_{n-2} C_1 + C_{n-1} C_0 = \sum_{k=0}^{n-1} C_k C_{n-k-1}$$

The initial conditions are $C_0 = 1$ and $C_1 = 1$.

The sequence $\{C_n\}$ is the sequence of **Catalan Numbers**.

Summary

Counting with Recurrence Relations

- Counting Bitstrings – same recurrence for different problems
- The Tower of Hanoi – recurrence we can solve
- Counting Arrangments of Parantheses – recurrence hard to solve

Solving Linear Recurrence Relations

Section 8.2

Video 68: Linear Recurrence Relations

- Linear Homogeneous Recurrence Relations
- Solving Linear Homogeneous Recurrence Relations

Linear Homogeneous Recurrence Relations

Definition: A **linear homogeneous recurrence relation of degree** k **with constant coefficients** is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots.. + c_k\, a_{n-k}$$

where $c_1, c_2, \ldots., c_k$ are real numbers, and $c_k \neq 0$

By strong induction, a sequence satisfying such a recurrence relation is uniquely determined by the recurrence relation and the *k* initial conditions
$a_0 = C_1, a_0 = C_1, \ldots, a_{k-1} = C_{k-1}$.

Terminology explained

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots.. + c_k\, a_{n-k}$

- is **linear** because the right-hand side is a sum of the previous terms of the sequence each multiplied by a function of n.
- is **homogeneous** because no terms occur that are not multiples of the a_js.
- has **constant coefficients** $c_1, c_2, \ldots., c_k$.
- the **degree** is k because a_n is expressed in terms of the previous k terms of the sequence.

Examples

$P_n = (1.11)P_{n-1}$	linear homogeneous recurrence relation of degree one
$f_n = f_{n-1} + f_{n-2}$	linear homogeneous recurrence relation of degree two
$a_n = a_{n-1} + a_{n-2}^2$	not linear
$H_n = 2H_{n-1} + 1$	not homogeneous
$B_n = nB_{n-1}$	coefficients are not constants

Characteristic Equation

Given the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k\, a_{n-k}$ assume $a_n = r^n$, where r is a constant

Substituting into the recurrence relation gives

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k\, r^{n-k}$$

Algebraic manipulation yields the **characteristic equation**:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_{k-1} r - c_k = 0$$

Solving Linear Homogeneous Recurrence Relations

The sequence $\{a_n\}$ with $a_n = r^n$ is a solution if and only if r is a solution to the characteristic equation.

- The solutions to the characteristic equation are called the **characteristic roots** of the recurrence relation.
- The roots can be used to give an **closed formula** for the recurrence relation.

Solving Linear Homogeneous Recurrence Relations of Degree Two

Theorem 1: Let c_1 and c_2 be real numbers. Suppose that

$$r^2 - c_1 r - c_2 = 0$$

has two distinct roots r_1 and r_2. Then the sequence $\{an\}$ is a solution to the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if

$$a_n = \alpha r_1^n + \alpha_2 r_2^n$$

for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

Example

What is the solution to the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$?

The characteristic equation is $r^2 - r - 2 = 0$.

Its roots are $r = 2$ and $r = -1$.

Therefore, $\{a_n\}$ is a solution to the recurrence relation if and only if $a_n = \alpha_1 2^n + \alpha_2(-1)^n$, for some constants α_1 and α_2.

To find the constants α_1 and α_2, note that

$$a_0 = 2 = \alpha_1 + \alpha_2 \text{ and } a_1 = 7 = \alpha_1 2 + \alpha_2(-1).$$

Solving these equations, we find that $\alpha_1 = 3$ and $\alpha_2 = -1$.

Hence, the solution is the sequence $\{a_n\}$ with $a_n = 3 \cdot 2^n - (-1)^n$.

Example: Fibonacci Numbers

The sequence of Fibonacci numbers satisfies the recurrence relation

$f_n = f_{n-1} + f_{n-2}$ with the initial conditions: $f_0 = 0$ and $f_1 = 1$.

The characteristic equation is $r^2 - r - 1$

And it's roots are $r_1 = \frac{1+\sqrt{5}}{2} \qquad r_2 = \frac{1-\sqrt{5}}{2}$

Therefore $f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$

Using the initial conditions $f_0 = 0$ and $f_1 = 1$, we have

$$f_0 = \alpha_1 + \alpha_2 = 0$$

$$f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right) = 1$$

Solving for α_1 and α_2 $\alpha_1 = \frac{1}{\sqrt{5}} \qquad \alpha_2 = -\frac{1}{\sqrt{5}}$

Hence $f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n$

Summary

- Linear Homogeneous Recurrence Relations
  - Characteristic equation
  - Characteristic roots
- Solving Linear Homogeneous Recurrence Relations of degree 2 with Constant Coefficients

Video 69: General Linear Recurrence Relations

- Homogeneous Recurrence Relations with Repeated Root
- Linear Homogeneous Recurrence Relations of Arbitrary Degree

Solving Linear Homogeneous Recurrence Relations with Repeated Root

Theorem 2: Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that

$$r^2 - c_1 r - c_2 = 0$$

has one repeated root r_0. Then the sequence $\{a_n\}$ is a solution to the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if

$$a_n = \alpha r_0^n + \alpha_2 n r_0^n$$

for $n = 0, 1, 2, \ldots$, where α and α_2 are constants.

Example

What is the solution to the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with $a_0 = 1$ and $a_1 = 6$?

The characteristic equation is $r^2 - 6r + 9 = 0$.

The only root is $r = 3$.

Therefore, $\{a_n\}$ is a solution to the recurrence relation if and only if $a_n = \alpha_1 3^n + \alpha_2 n(3)^n$ where α_1 and α_2 are constants.

To find the constants α_1 and α_2, note that

$$a_0 = 1 = \alpha_1 \quad \text{and} \quad a_1 = 6 = \alpha_1 \cdot 3 + \alpha_2 \cdot 3.$$

Solving, we find that $\alpha_1 = 1$ and $\alpha_2 = 1$.

Hence, $a_n = 3^n + n3^n$.

Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

Theorem 3: Let $c_1, c_2, \ldots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \ldots - c_k = 0$$

has k distinct roots $r_1, r_2, \ldots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots\ldots + c_k a_{n-k}$$

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \cdots + \alpha_k r_k^n$$

for $n = 0, 1, 2, \ldots$, where $\alpha_1, \alpha_2, \ldots, \alpha_k$ are constants.

The General Case with Repeated Roots Allowed

Theorem 4: Let $c_1, c_2, \ldots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \cdots - c_k = 0$$

has t distinct roots $r_1, r_2, \ldots, r_t$ with multiplicities $m_1, m_2, \ldots, m_t$, respectively so that $m_i \geq 1$ for $i = 0, 1, 2, \ldots, t$ and $m_1 + m_2 + \ldots + m_t = k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots\ldots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n = {} & (\alpha_{1,0} + \alpha_{1,1} n + \cdots + \alpha_{1,m_1-1} n^{m_1-1}) r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1} n + \cdots + \alpha_{2,m_2-1} n^{m_2-1}) r_2^n \\ & + \cdots + (\alpha_{t,0} + \alpha_{t,1} n + \cdots + \alpha_{t,m_t-1} n^{m_t-1}) r_t^n \end{aligned}$$

for $n = 0, 1, 2, \ldots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_{i-1}$.

Example

What is the solution to the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with $a_0 = 1$, $a_1 = -2$, and $a_2 = -1$.

The characteristic equation is $r^3 + 3r^2 + 3r + 1 = 0$.

The only root is $r = -1$.

Therefore, $\{a_n\}$ is a solution to the recurrence relation if and only if $a_n = \alpha_{1,0}(-1)^n + \alpha_{1,1}n(-1)^n + \alpha_{1,2}n^2(-1)^n$

where $\alpha_{1,0}$, $\alpha_{1,1}$ and $\alpha_{1,2}$ are constants.

To find the constants $\alpha_{1,0}$, $\alpha_{1,1}$ and $\alpha_{1,2}$, note that

$a_0 = 1 = \alpha_{1,0}$ $\quad$ $a_1 = -2 = -\alpha_{1,0} - \alpha_{1,1} - \alpha_{1,1}$ $\quad$ $a_2 = -2 = \alpha_{1,0} + 2\alpha_{1,1} + 4\alpha_{1,1}$

Solving, we find that $\alpha_{1,0} = 1$, $\alpha_{1,1} = 3$ and $\alpha_{1,2} = -2$

Hence, $a_n = (1 + 3n - 2n^2)(-1)^n$

Summary

- Homogeneous Recurrence Relations with Repeated Root
- Linear Homogeneous Recurrence Relations of Arbitrary Degree

Generating Functions

Section 8.4

Video 70: Generating Functions

- Generating Functions
- Solving recurrence relations with generating functions

Generating Functions

Definition: The **generating function for the infinite sequence** $a_0, a_1, \ldots, a_k, \ldots$ of real numbers is the infinite series

$$G(x) = a_0 + a_1 x + \cdots + a_k x^k + \cdots = \sum_{k=0}^{\infty} a_k x^k.$$

Useful Generating Functions

TABLE 1 Useful Generating Functions.

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^{n} C(n,k)x^k = 1 + C(n,1)x + C(n,2)x^2 + \cdots + x^n$	$C(n,k)$
$(1+ax)^n = \sum_{k=0}^{n} C(n,k)a^k x^k = 1 + C(n,1)ax + C(n,2)a^2x^2 + \cdots + a^n x^n$	$C(n,k)a^k$
$(1+x^r)^n = \sum_{k=0}^{n} C(n,k)x^{rk} = 1 + C(n,1)x^r + C(n,2)x^{2r} + \cdots + x^{rn}$	$C(n,k/r)$ if $r \mid k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^{n} x^k = 1 + x + x^2 + \cdots + x^n$	1 if $k \le n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2x^2 + \cdots$	a^k
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \cdots$	1 if $r \mid k$; 0 otherwise
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \cdots$	$k+1$
$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)x^k = 1 + C(n,1)x + C(n+1,2)x^2 + \cdots$	$C(n+k-1,k) = C(n+k-1,n-1)$
$\frac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)(-1)^k x^k = 1 - C(n,1)x + C(n+1,2)x^2 - \cdots$	$(-1)^k C(n+k-1,k) = (-1)^k C(n+k-1,n-1)$
$\frac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1,k)a^k x^k = 1 + C(n,1)ax + C(n+1,2)a^2x^2 + \cdots$	$C(n+k-1,k)a^k = C(n+k-1,n-1)a^k$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$1/k!$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$(-1)^{k+1}/k$

Note: The series for the last two generating functions can be found in most calculus books when power series are discussed.

Examples

The sequence $\{a_k\}$ with $a_k = 3$ has the generating function $\sum_{k=0}^{\infty} 3x^k.$

The sequence $\{a_k\}$ with $a_k = k + 1$ has the generating function has the generating function

$$\sum_{k=0}^{\infty} (k+1)x^k.$$

The sequence $\{a_k\}$ with $a_k = 2^k$ has the generating function has the generating function

$$\sum_{k=0}^{\infty} 2^k x^k.$$

Solving Recurrence Relations with Generating Functions

Solve the recurrence relation $a_k = 3a_{k-1}$ with initial condition $a_0 = 2$.

Take the generating function , $G(x) = \sum_{k=0}^{\infty} a_k x^k$.

Observe that

$$xG(x) = \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k.$$

Now we see that

$$\begin{aligned} G(x) - 3xG(x) &= \sum_{k=0}^{\infty} a_k x^k - 3 \sum_{k=1}^{\infty} a_{k-1} x^k \\ &= a_0 + \sum_{k=1}^{\infty} (a_k - 3a_{k-1}) x^k \\ &= 2, \end{aligned}$$

Solving Recurrence Relations with Generating Functions

Thus $G(x) - 3xG(x) = (1 - 3x)G(x) = 2.$

or $G(x) = 2/(1 - 3x)$

Using $1/(1 - ax) = \sum_{k=0}^{\infty} a^k x^k$

We get $G(x) = 2\sum_{k=0}^{\infty} 3^k x^k = \sum_{k=0}^{\infty} 2 \cdot 3^k x^k.$

And thus $a_k = 2 \cdot 3^k$

Solving Recurrence Relations with Generating Functions

Given some recurrence relation for a sequence a_0, a_1,…, a_k, …

General Approach to find a closed formula

- Interpret the sequence as coefficients of a generating function G(x)
- Use the recurrence relation to derive an alternative expression for G(x)
- Determine the power expansion of this alternative expression of which the coefficients must be equal to the sequence

Solving Hanoi Tower

Solving the towers of Hanoi with a generating function Let $M_0 = 0$ and $M_i = 2M_{i-1} + 1$ for $i > 0$. Put $H(x) = \sum_{i=0}^{\infty} M_i x^i$, then

$$\begin{aligned} H(x) &= \sum_{i=1}^{\infty} 2M_{i-1}x^i + \sum_{i=1}^{\infty} x^i \\ &= 2\sum_{j=0}^{\infty} M_j x^{j+1} + x\sum_{i=0}^{\infty} x^i \\ &= 2x\sum_{j=0}^{\infty} M_j x^j + x\sum_{i=0}^{\infty} x^i \\ &= 2xH(x) + \frac{x}{1-x}. \end{aligned}$$

This implies that $H(x) = \frac{x}{(1-x)(1-2x)}$. Solving $H(x) = \frac{u}{1-x} + \frac{v}{1-2x}$ for u and v it is found that $u - 2ux + v - vx = x$ and thus $u = -v$ and $-2u - v = -2u + u = 1$ so that $u = -1$ and $v = 1$. It follows that $H(x) = \frac{1}{1-2x} - \frac{1}{1-x}$, therefore that

$$H(x) = \sum_{i=0}^{\infty} (2x)^i - \sum_{i=0}^{\infty} x^i,$$

and thus $M_n = 2^n - 1$.

Summary

- Generating Functions
- Useful generating functions
- Solving recurrence relations with generating functions

Video 71: Counting Problems

- Solving counting problems with generating functions

Counting Problems and Generating Functions

Find the number of solutions of

$$e_1 + e_2 + e_3 = 17,$$

where e_1, e_2, and e_3 are nonnegative integers with

$$2 \le e_1 \le 5,\ 3 \le e_2 \le 6, \text{ and } 4 \le e_3 \le 7.$$

The number of solutions is the coefficient of x^{17} in the expansion of

$$f(x) = (x^2 + x^3 + x^4 + x^5)\,(x^3 + x^4 + x^5 + x^6)\,(x^4 + x^5 + x^6 + x^7).$$

A term equal to x^{17} is obtained in the product by picking
x^{e_1} in the first sum, x^{e_2} in the second sum x^{e_2}, and x^{e_3} in the third sum x^{e_3},
such that $e_1 + e_2 + e_3 = 17$.

There are three solutions since the coefficient of x^{17} in the product is 3.

Combinations

Find the number of *k*-combinations of a set with n elements using generating functions

- The number of *k*-combinations is the coefficient of x^k in the generating function

$$f(x) = (1+x)^n$$

- Using the binomial theorem: $f(x) = \sum_{k=0}^{n} \binom{n}{k} x^k$
- Therefore the number of k-combinations is $\binom{n}{k}$

Extended Binomial Coefficients

Definition: Let u be a real number and k a nonnegative integer. Then the extended binomial coefficient $\binom{u}{k}$ is defined as

$$\binom{u}{k} = \begin{cases} \dfrac{u(u-1)\cdots(u-k+1)}{k!}, if\ k > 0 \\ 1, if\ k = 0 \end{cases}$$

Example:

$$\binom{-2}{3} = \frac{-2\cdot -3\cdot -4}{3!} = -4 \qquad \binom{1/2}{3} = \frac{1/2\cdot(\frac{1}{2}-1)\cdot(\frac{1}{2}-2)}{3!} = \frac{1}{16}$$

Extended Binomial Theorem

Theorem: Let x be real number with |x|<1 and let u be real number. Then

$$(1+x)^u = \sum_{k=0}^{n} \binom{u}{k} x^k$$

Combinations with Repetition

Find the number of *k*-combinations of a set with n elements using generating functions when repetition is allowed

- The number of *k*-combinations is the coefficient of x^k in the generating function
$$f(x) = (1 + x + x^2 + \cdots)^n$$
- As long as |x|< 1 we have $(1 + x + x^2 + \cdots) = \frac{1}{1-x}$
- Therefore $f(x) = \frac{1}{(1-x)^n} = (1 + (-x))^{-n} = \sum_{k=0}^{n} \binom{-n}{k} x^k$

Combinations with Repetition

- The coefficient of x^k is $\binom{-n}{k}(-1)^k$

$$\binom{-n}{k} = \frac{(-n)(-n-1)\dots(-n-k+1)}{k!} =$$
$$\frac{(-1)^k(n)(n+1)\dots(n+k-1)}{k!} = \frac{(-1)^k(n+k-1)\dots(n)}{k!} =$$
$$\frac{(-1)^k(n+k-1)!}{k!(n-1)!} = (-1)^k\binom{n+k-1}{k}$$

- Therefore $\binom{-n}{k}(-1)^k = \binom{n+k-1}{k}$

Summary

- Counting combinations with generating functions
- Extended Binomial Theorem
- Counting Combinations with Repetition with generating functions

Inclusion-Exclusion

Section 8.5

Video 72: Inclusion-Exclusion

- The Principle of Inclusion-Exclusion
- Examples

Principle of Inclusion-Exclusion

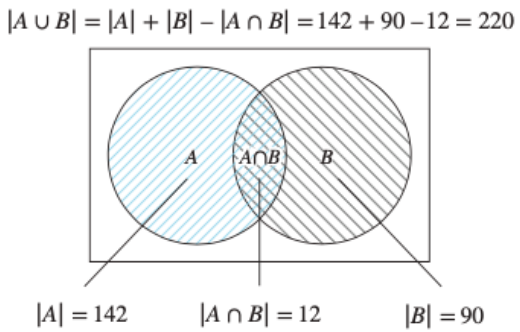
We have shown that for finite sets A and B

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example: How many positive integers less or equal 1000 are divisible by 7 or 11?

For integer n are $\left\lfloor \frac{1000}{n} \right\rfloor$ integers less or equal 1000 divisible by n

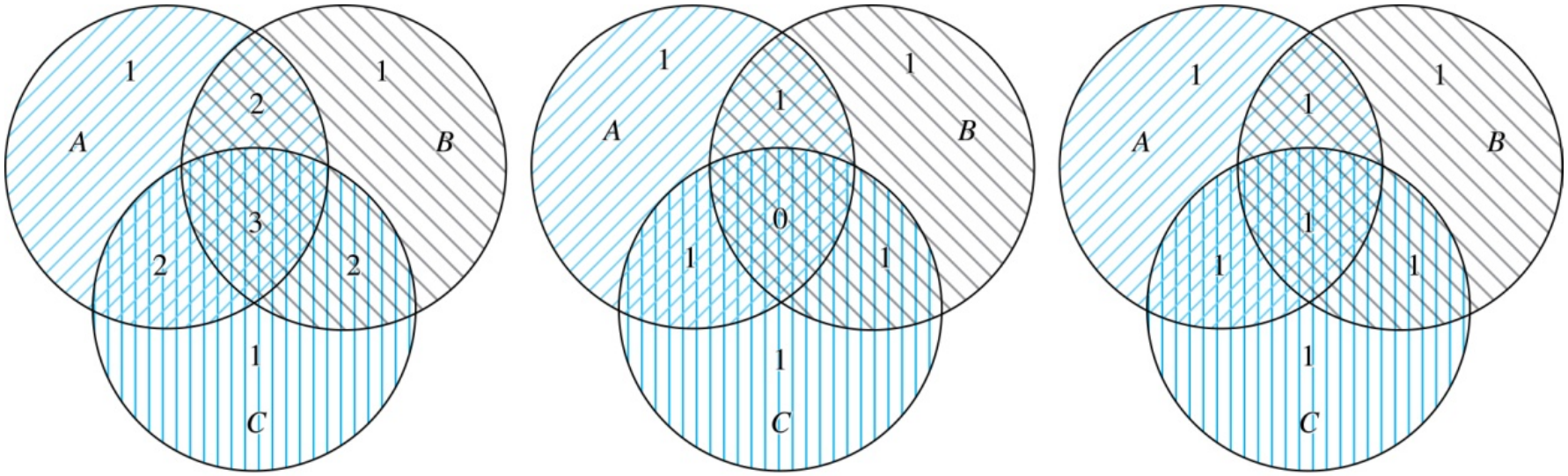
Therefore $\left\lfloor \frac{1000}{7} \right\rfloor + \left\lfloor \frac{1000}{11} \right\rfloor - \left\lfloor \frac{1000}{7 \cdot 11} \right\rfloor = 142 + 90 - 12 = 220$ integers less or equal 1000 are divisible by 7 or 11.



Three Finite Sets

$$|A \cup B \cup C| =$$

$$|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



(a) Count of elements by $|A|+|B|+|C|$

(b) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|$

(c) Count of elements by $|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|+|A\cap B\cap C|$

Example

How many positive integers less or equal 1000 are divisible by 5, 7, or 11?

$$\left\lfloor\frac{1000}{5}\right\rfloor + \left\lfloor\frac{1000}{7}\right\rfloor + \left\lfloor\frac{1000}{11}\right\rfloor - \left\lfloor\frac{1000}{5\cdot 7}\right\rfloor - \left\lfloor\frac{1000}{5\cdot 11}\right\rfloor - \left\lfloor\frac{1000}{7\cdot 11}\right\rfloor + \left\lfloor\frac{1000}{5\cdot 7\cdot 11}\right\rfloor =$$

$$= 200 + 142 + 90 - 28 - 18 - 12 + 2 = 372$$

integers less or equal 1000 are divisible by 5, 7, or 11

The Principle of Inclusion-Exclusion

Theorem 1. The Principle of Inclusion-Exclusion: Let $A_1, A_2, \ldots, A_n$ be finite sets. Then:

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = \sum_{1 \le i \le n} |A_i| - \sum_{1 \le i < k \le n} |A_i \cap A_k| +$$

$$+ \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| - \ldots + (-1)^n \left| \bigcap_{1 \le i \le n} A_i \right|$$

The Principle of Inclusion-Exclusion

Proof: Consider an element a that is a member of r of the sets $A_1, \ldots, A_n$ where $1 \leq r \leq n$.

- It is counted $C(r, 1)$ times by $\sum |A_i|$
- It is counted $C(r, 2)$ times by $\sum |A_i \cap A_j|$
- In general, it is counted $C(r, m)$ times by the summation of m of the sets A_i.

The Principle of Inclusion-Exclusion

Thus the element is counted exactly

$$C(r,1) - C(r,2) + C(r,3) - \cdots + (-1)^{r+1} C(r,r)$$

times by the right hand side of the equation.

Using the binomial theorem

$$\sum_{k=0}^{n} C(r,k)(-1)^k = \sum_{k=0}^{n} \binom{r}{k} (-1)^k = (1+(-1))^n = 0$$

we obtain

$$C(r,0) - C(r,1) + C(r,2) - \cdots + (-1)^r C(r,r) = 0.$$

Hence,

$$1 = C(r,0) = C(r,1) - C(r,2) + \cdots + (-1)^{r+1} C(r,r).$$

Derangements

Definition: A **derangement** is a permutation of objects that leaves no object in the original position.

Example:

The permutation of 21453 is a derangement of 12345.

But 21543 is not a derangement of 12345, because 4 is in its original position.

Derangements

Theorem 2: The number of derangements of a set with *n* elements is

$$D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right].$$

Derangements - Proof

Proof: Let P_i be the set of permutations that fix element i.

The set of permutations that are not derangements is then $P_1 \cup P_2 \cup \ldots \cup P_n$.

Thus the number of derangements $D_n = n! - |P_1 \cup P_2 \cup \ldots \cup P_n|$.

Now $|P_i| = (n-1)!$, $|P_i \cap P_k| = (n-2)!$, if $i \neq k$, and in general $\left|\bigcap_{i \in I} P_i\right| = (n-s)!$ if $|I| = s$.

Using the principle of inclusion-exclusion:

$|P_1 \cup P_2 \cup \ldots \cup P_n| =$

$$\sum_{1 \le i \le n} |P_i| - \sum_{1 \le i < k \le n} |P_i \cap P_k| + \cdots + (-1)^n \left|\bigcap_{1 \le i \le n} P_i\right| = \binom{n}{1}(n-1)! - \binom{n}{2}(n-2)! + \cdots + \binom{n}{n}(-1)^n$$

and since $\binom{n}{k}(n-k)! = \frac{n!}{k!}$

and therefore $D_n = n! - \frac{n!}{1!} + \frac{n!}{2!} - \cdots + \frac{n!}{n!}(-1)^n = n!\left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots - (-1)^n \frac{1}{n!}\right)$

Example

The Hatcheck Problem: A new employee checks the hats of *n* people at restaurant, forgetting to put claim check numbers on the hats.

When customers return for their hats, the checker gives them back hats chosen at random from the remaining hats.

What is the chance that no one receives the correct hat?

The number of ways the hats can be arranged so that there is no hat in its original position divided by *n*!, the number of permutations of *n* hats.

$$\frac{D_n}{n!} = \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!}\right]$$

TABLE 1 **The Probability of a Derangement.**

n	2	3	4	5	6	7
$D_n/n!$	0.50000	0.33333	0.37500	0.36667	0.36806	0.36786

Summary

- Principle of Inclusion-Exclusion for 2 sets
- Principle of Inclusion-Exclusion for 3 sets
- Principle of Inclusion-Exclusion for n sets
- Number of Derangements